AYUSH AGGARWAL

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SUMMARY

Full-stack engineer, backend-leaning, with forward-deployed-style experience (onboarding, UAT, data integration) for college and pharma clients. 4+ years building **AWS backend, data and platform systems** at Ellucian and ZS. Open to roles across the US.

WORK EXPERIENCE

Ellucian · Insights Platform, Reporting & Analytics — Jun 2024 - Present

Software Engineer II (Jun 2026 - Present) · Software Engineer (Jun 2024 - Jun 2026)

- Owned the **Metabase-Athena lakehouse integration** end to end: worked with the Metabase team to resolve JDBC and metadata-sync issues, and implemented **database routing** (a router database plus hidden destination databases with separate IAM credentials, chosen per user by a login attribute).
- Onboarded customers onto Metabase reporting and **led user acceptance testing (UAT) with dozens of college users**.
- Co-designed **role-based lakehouse data access** with AWS SMEs: a login resolver that unions a user's app roles into a permission package, mapped to IAM credentials and Lake Formation grants, DynamoDB tracking, SSM credentials.
- Built backend services (Node.js, AWS Lambda) for Ellucian's **agentic chat interface**, now shipped to customers: a permission-aware layer that calls existing reporting APIs, resolves each user's roles through the lakehouse routing and access model, and returns structured results the agent turns into conversational insights.
- Identified long-lived IAM credentials as a security and maintenance risk, won alignment from senior architects, and built an **automated credential-rotation service** (Jenkins, DynamoDB metadata, scheduler, validation) that is deployed and running.
- Led the **Node.js 22 upgrade across 40+ services** (runtime, dependency and CI compatibility), with automated validation.
- Contributed to the **streaming ingestion platform** (Python, Kafka/MSK, Flink): shipped DII transformation of nested documents into relational tables, added _document_key indexes on child tables, and built a source-type allowlist with a **CI validation gate** that fails builds on unsupported config.
- Built a **Lambda-based Datadog log-triage pipeline** that cut manual operational effort by **50%**, and shipped accessible (WCAG, right-to-left) React reporting features.

Ellucian · Software Engineer Intern — May 2023 - Aug 2023

- Built a self-service **Insights Card** template for Metabase embeds (React, Lambda, Terraform), supporting **500+ assets**.

ZS Associates · Software Engineer · Pharma client work · Gurgaon, India — Aug 2020 - Jun 2022

- On a global pharma transparency platform, **harmonized data from ~50 countries** (differing formats and languages) into one model using per-country mapping and validation rules (Python, SQL on Athena/Redshift, Parquet on S3), giving client teams a single global view for KPI dashboards and cutting the time users spent on KPI reporting and manual reconciliation.
- Migrated legacy **MySQL/Oracle pipelines to AWS** (S3, Athena, Redshift), improving query performance **~30%** and integration throughput **~20%** with partitioning and Parquet; built data apps and APIs that pushed modeled data into downstream tools, with orchestration, retries and monitoring.
- Worked directly with **dozens of client users and stakeholders**: trained them on the dashboards, troubleshot data issues they reported as production support, and kept scheduled syncs reliable.

PROJECTS

The Couch *(Next.js, TypeScript, Supabase)* | roomies-vert.vercel.app | github.com/ayush51/social-rommies-hub

- Private household PWA where access is enforced in Postgres row-level security, not app code; an append-only ledger with idempotency keys means retries can't double-pay.
- Realtime chat, web push, and server-validated chess and Monopoly-style games; Playwright suite across 5 screen sizes.

StockLens *(Next.js, Prisma, Postgres)* | thesistrack-eight.vercel.app | github.com/ayush51/stocklens

- Research dashboard tying portfolios to investment theses; brokerage access is read-only by design, enforced by types.
- Signal scoring that cross-references SEC EDGAR insider filings with USAspending.gov contract awards, with entity resolution.

SideQuest *(Next.js, Supabase)* | sidequestdmv.vercel.app | github.com/ayush51/sidequest

- Mobile-first PWA that hands out a daily real-world quest, with points, streaks and friends; user-suggested quests pass an admin approval queue, and RLS limits what friends see.

Sip Happens *(Next.js, Gemini API, Supabase)* | drink-tracker-lime.vercel.app | github.com/ayush51/drink-tracker

- Drink tracker where a vision model (Gemini) estimates a drink from a photo; daily limits, history, group leaderboard and web push reminders in an installable PWA.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, Java, C/C++, Bash · Frontend: React, Next.js, HTML/CSS
Backend & Data:	Node.js / Express, REST APIs, Postgres, data pipelines, Metabase, Athena
Cloud:	AWS (Lambda, S3, IAM, Lake Formation, DynamoDB, SQS/SNS, SSM), Terraform, Docker
Tools:	Git, Jenkins (CI/CD), Datadog, Playwright, pytest

EDUCATION & RECOGNITION

Virginia Tech, M.Eng, Computer Science · GPA 3.96/4 · Graduate Research Assistant — Aug 2022 - May 2024

GGSIP University, Delhi, B.Tech, Information Technology · GPA 9.12/10 · University first rank (2017) — Aug 2016 - Jul 2020

Awards at Ellucian: “Impact and Recognition” Award (client impact); “Celebrating Impact - Best Team” Award.